

EE3TR4 Lab 1: Fourier Analysis

Please work in pairs. *Please note that each group is expected to work independently.* Please be familiar with the guidelines on academic integrity, summarized on the course outline available on the web. The late penalty is 10% per day.

This lab is project based, consisting of two parts. The first part involves Fourier analysis. Here, you calculate the output of a filter of your choice due to a square wave input. The second part is less intensive and involves listening to various forms of filtered speech, to give you an idea what the effect of filtering operations are on audio signals.

You can attend any lab session you wish. However, please note that if you show up at a lab session that is not your regularly scheduled one, then those who are assigned to that session have priority. You may also do your labs during off-hours if you wish, but then TAs won't be on hand to help. You will need to validate your ID card at the hub, JHE 216A, to gain access after regular hours.

Part I: Filter Analysis

This is an open-ended project, and so details are purposely left vague. Design and implement a filter of your choice, and apply a square wave as input. Calculate the corresponding (periodic) output waveform. Compare your calculated waveforms with the measurements you obtain in the lab. Investigate the effect of varying the values of parameters such as cutoff frequency of the filter and the period of the input signal. We expect a report that is clear, articulate and accurate.

Here is a list of various components you will have available in the lab:

Op Amps

0.001, 0.01 and 0.1 μ F capacitors

Resistors: 470, 560, 680, 820, 1K, 1.2K, 1.8K, 2.0K, 2.4K, 3.3K, 3.9K, 4.7K, 5.6K, 6.2K, 8.2K, 10K.

Hints and Suggestions:

1. You can calculate the output response $y(t)$ of the filter in at least two different ways-- in the frequency domain and in the time domain. For the frequency domain, we have $Y(nf_0) = X(nf_0)H(nf_0)$, where f_0 is the fundamental frequency of the periodic input signal. The spectrum $X(nf_0)$ is known, since it can be calculated from the Fourier transform of the input signal. The filter frequency response $H(f)$ can be calculated knowing the transfer function of the filter, which is available from the supplementary material for this lab. Thus, $Y(nf_0)$ can be determined. The corresponding time-domain

signal $y(t)$ can be found by inverting the Fourier transform quantity $Y(f)$, as we discussed in class. Note that the transfer functions of the filters shown in the supplementary material are normalized to 1 radian/sec. To achieve a cutoff frequency of ω_c radians/sec., replace ω with the quantity $\frac{\omega}{\omega_c}$. Instead of Fourier transform, you can also use Fourier series representation as done in Tutorial 1 or in the matlab code “Triangular_filtering.m”.

The output can also be determined in the time domain as follows. We know that $y(t) = h(t) \otimes x(t)$, where $\otimes$ denotes convolution. The input signal $x(t)$ is known. The impulse response $h(t)$ can be determined as the inverse FFT of the frequency response $H(f)$. This can easily be done using the matlab command “ifft”. There are a few tricks to be aware of in this operation, so be careful. See point 8 below.

2. Design your filter circuit using the components available before you go into the lab.
 3. The spectrum of any signal can be measured with the scopes, using the FFT function. A scope manual is available as an icon on the desktop of the lab computers.
 4. The Butterworth lowpass active filter configuration is probably the most straightforward. You can use a 1st-, 2nd-, 3rd- or 4th-order filter, as you see fit. See the material on active filter design that has been posted on the website.
 5. Make sure your cutoff frequency is a reasonable choice, relative to the frequency of your input signal. You don't want the cutoff frequency of the filter to be so large that few frequency components of the input are removed by the filter.
 6. You can measure the frequency response of your filter if you wish, by applying an input sinusoid at varying frequencies, and measuring the corresponding relative amplitude and phase of the output.
 7. Try varying the period (or equivalently fundamental frequency) of the input wave. Comment on its effect on the output waveform.
 8. To evaluate a discrete-time version of the impulse response $h(nT_s)$ of your filter, assemble a vector in Matlab of values of $H(j2\pi f)$ according to the format of the Discrete Fourier Transform. Make sure the sampling frequency f_s is large enough so that $|H(\frac{f_s}{2})|$ is small (e.g. 0.01) in comparison to $|H(0)|$ to avoid aliasing error in frequency, and that N (the number of samples) is large enough so that $h(NT_s)$ is small compared to its largest magnitude, to avoid aliasing in time.
- used series representation
- If you use Fourier series representation, you don't need to worry about Discrete Fourier Transform. Simply multiply the Fourier coefficient of input by the filter transfer function to get the Fourier coefficient of the output and then sum up the Fourier series to get the time domain signal.
9. A 2nd-order or larger filter is more interesting than a first order.

Part 2: Speech Filtering

This should be a bit more fun than part I. On the computers in the lab, run the matlab program "filtDemo". It plays speech that has been passed through various forms of filter. Plots of the input and output spectra are shown on the displays. Note especially how the notch filter can suppress the sinusoidal tone. Pay attention to the difference in sound of the outputs of each type of filter.

put lab code

input → triang → 

coeff. tri → 

trans func : 1st order → 2nd order

Lab 1: "Fourier Analysis"

UA 741 PC

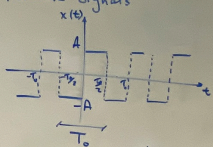
Input/output Relation

FS representation of Periodic Signals

→ I/P: $x(t)$

$$f_0 = \frac{1}{T_0}$$

500
1000
1500
2000
Fund. Freq.



→ Complex FSC:

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{j\frac{2\pi n}{T_0}t}$$

where

$$C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-j\frac{2\pi n}{T_0}t} dt$$

*for the shown signal with

$$C_n = A e^{j\frac{(n\pi)^2}{2n\pi}} = \frac{-jA [1 - (-1)^n]}{n\pi} = \begin{cases} \frac{2A}{n\pi} & n = \text{odd} \\ 0 & n = \text{even} \end{cases}$$

$$x(t) \rightarrow [h(t)] \rightarrow y(t)$$

$$y(t) = x(t) * h(t)$$

$$Y(f) = X(f) \cdot H(f)$$

*for periodic signals

$$Y(nf_0) = X(nf_0) \cdot H(nf_0)$$

$$C_n^y = C_n^x H(nf_0)$$

$$y(t) = \sum_{n=-\infty}^{\infty} C_n^y e^{j\frac{2\pi n}{T_0}t}$$

Butterworth filter Design

$$A_v(s) = \frac{A_v}{B_n(s)}$$

for a LPF with 3-dB point @ ω_c

$$|A_v(s)|^2 = \frac{A_v^2}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}$$

1 kHz
5 kHz
10 kHz
20 kHz
30 kHz

BW. Poly. of degree (n)

$$H_c(f) = \frac{A_v \cdot 1.6}{\left(\left(\frac{f}{f_c}\right)^2 + 2K\left(\frac{f}{f_c}\right) + 1\right)^{n/2}}$$

Ex: A 2nd Order LPF

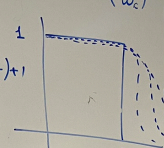
$$B_n(s) = \left(\frac{s}{\omega_c}\right)^2 + 2K\left(\frac{s}{\omega_c}\right) + 1$$

where

$$K = \dots = 0.707 \rightarrow \text{from the Table}$$

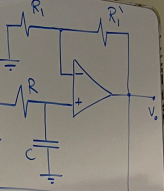
$$A_v = \frac{R_1 + R_1}{R_1} = 3 - 2K$$

$$\omega_c = \frac{1}{RC}$$



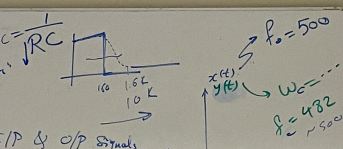
Steps:

- 1) Find K from the table (ex: K=0.707 for n=2)
- 2) Cal. $A_v = 3 - 2K$ (AE=1.6 for n=2)
- 3) Choose $R \parallel R_1$ to get $A_v = \frac{R_1 + R}{R_1}$
- 4) " $R \parallel RC$ " $\omega_c = \frac{1}{RC}$



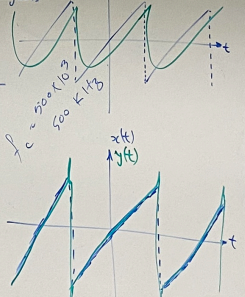
Before you start:

- 1) Download & run "triangular-fitting.m"
- 2) Note the last figure "Figure 5" for a comparison between the I/P & O/P signals



- 3) Change R from 3.3e3 to 3.3
- 4) Run the code again and note "Figure 5"

5) Comment !!



Lab

filter

Work in pairs for both the lab and the pre-lab portions. Submit one report per group, on Avenue. Late penalty is 10% per week.

In this lab we get to look at the waveforms involved in double sideband suppressed carrier (DSB/SC) modulation systems, for a sinusoidal message signal. We will use special circuit boards that form a modulated DSB/SC carrier signal from a 64 KHz sinusoidal carrier and a 4KHz sinusoidal message signal. A block diagram of the circuit board is shown in Figure 1. The fully detailed schematic is shown in Figure 2. Note that the 4 kHz message is derived from the 64kHz carrier by converting the sinusoidal carrier to a square wave, dividing the frequency by 16, and then low-pass filtering the resulting square-wave output to generate the 4kHz sinusoidal message. The modulated waveform is also demodulated to recover the original message, by multiplying the modulated DSB/SC signal by the same 64 KHz carrier.

In this lab, we analyse signals in the time and frequency domains. Thus, we use the frequency domain display on the scopes. If you do not know how to use this feature, consult the scope manual or ask the TAs for assistance.

Note that this circuitry is not ideal. The multipliers are quite sensitive to variations and imbalances in the surrounding resistor values. Due to temperature shifts and aging, these variations are inevitable. As a result, the waveforms you observe will appear somewhat different from what you expect. But this behaviour is part of the life of an electronic engineer-- electronic circuitry is never ideal.

Pre-Lab Exercises

1. Learn your material on DSB/SC modulation.
2. Write a matlab program which gives the following outputs:
 - a) A plot of $s(t)$ corresponding to a DSB/SC waveform for a 64 KHz carrier and a 4 KHz sinusoidal message, in both the time and frequency domains.
 - b) Plot the DSB/SC signal in both the time and frequency domains when a carrier component is added to the signal. Consider the following cases: 1) The carrier power is 50% of the total power in the two sidebands, 2) the carrier power is 3 times the total power in the two sidebands, and 3) the carrier power is more than 3 times the power in the sidebands.
 - c) A plot of the resulting signal, in both the time and frequency domains, when $s(t)$ (with no carrier) is again multiplied by $\cos(2\pi f_c t)$.
3. For each of the above cases in part 2, give the underlying theoretical development.
4. Study the circuit diagrams (especially Figure 1) so that you can understand how the signals are generated, and know what the waveforms should look like at the various test points. You should also have a feeling what happens when each of the potentiometers are varied.

1. Transmit Section

Please note the following terminology that is used in the following:

TP means “test point”. This is a point on the circuit board you can place your scope probe.

PT means “potentiometer”. These can be varied to create various effects. Refer to the circuit diagrams in Figs. 1 and 2 to see where these are located.

Set the signal generator to a 64 kHz, ± 10 Volt sinusoidal waveform. Apply the signal generator and the power supply to the board. The message signal $m(t)$ appears at test point 2 (TP2). The multiplier output (i.e., the DSBSC signal $s(t)$) appears at TP3. Observe $s(t)$ in both the time and frequency domains on the scope. Use the Hanning window function for the frequency domain display. Note that the scope display is not ideal— an impulse appears as a lobe with a finite width.

scope
↓
 $m(t) \rightarrow 1$ yellow
 $s(t) \rightarrow 2$ green

- Write a brief description comparing your observed DSBSC waveform from the scope and the matlab output, in both the time and frequency domains. Give a brief explanation of the theory. If a printer is available in the lab, print the observed waveform from the scope. Otherwise, don't try to copy the observed waveforms to paper manually.
- By varying potentiometer PT1, test the DSBSC signal for linearity. Compare with your theoretical predictions.
- Ideally, the DSBSC signal contains no carrier component. But, inevitably, some carrier component will always leak through the multiplier. The carrier component of the DSBSC wave can be varied by adjusting PT2. Vary this potentiometer and observe the effect on the signal in both the time and frequency domains. Write a brief description comparing your observed time and frequency domain signals with your matlab plots for a DSBSC wave with carrier added. Support your description with appropriate theoretical analysis.
- From the perspective gleaned above, what is the difference between a sinusoidally modulated DSBSC wave, and a 100% sinusoidally modulated AM wave. (You will note that superficially, they appear almost the same!).
- Explain how the 4KHz message signal is generated by the board. What is the advantage of generating it this way?

2. Receive Section

Here, the DSBSC signal is multiplied again in the time domain by exactly the same carrier. The second multiplier output appears at TP4, and the final demodulated output appears at TP5. The waveform at TP4 may have to be adjusted using PT3 so that it more closely agrees with your matlab predictions.

- Again, write a brief description comparing the observed waveform at TP4 with that obtained from matlab, in both the time and frequency domains. You will note the observed waveform has a somewhat unusual shape. Support your descriptions with theoretical analysis.
- What operation is performed on the second (receiver) multiplier output to extract the final demodulated signal? Explain how this operation is performed on our circuit boards.
- Comment on the similarity of the demodulated output to the original message. What can be done to

improve the similarity?

d) By varying PT1, test the linearity of the overall modulator/demodulator section.

64 kHz $\pm 10V$

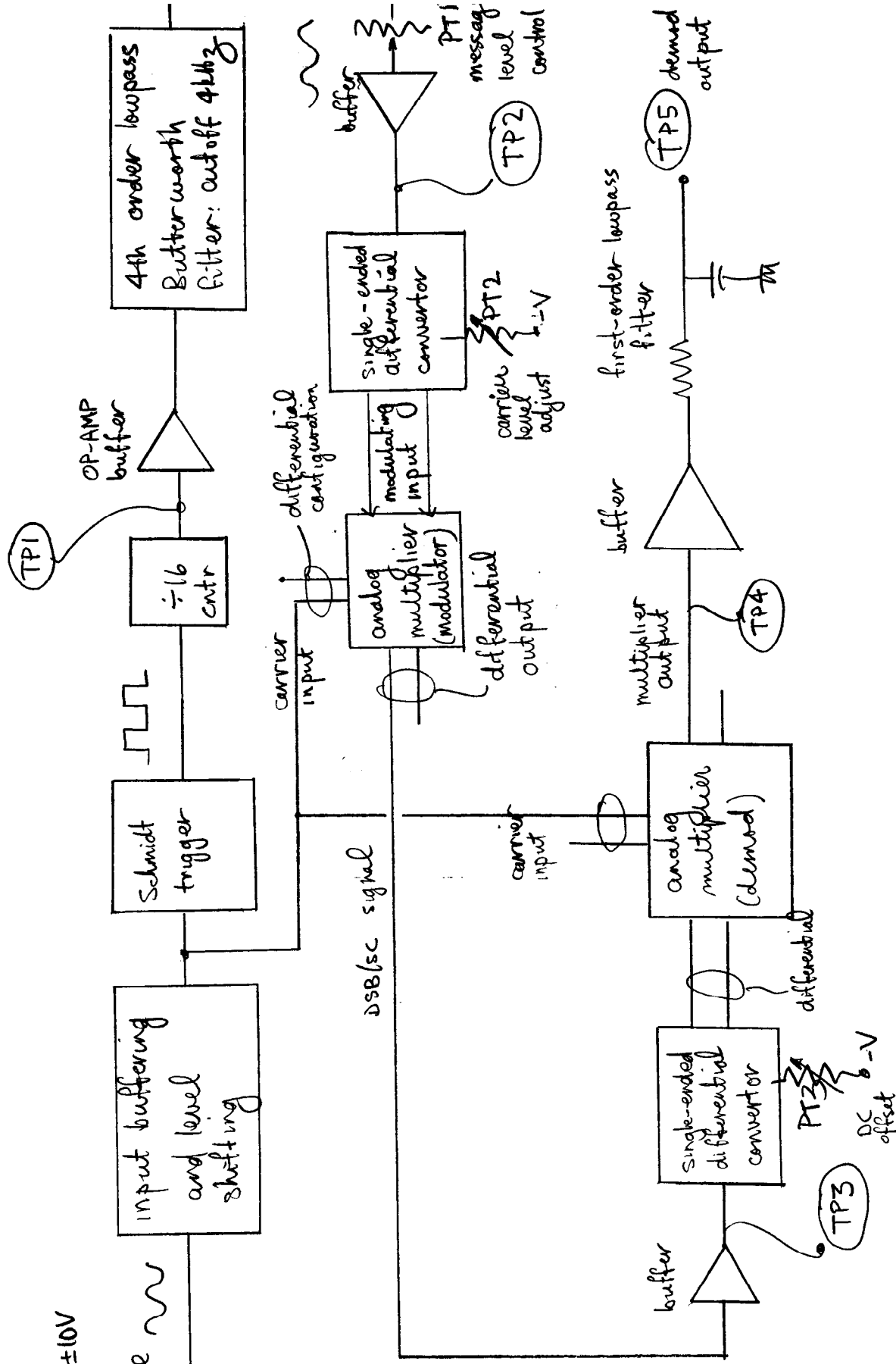
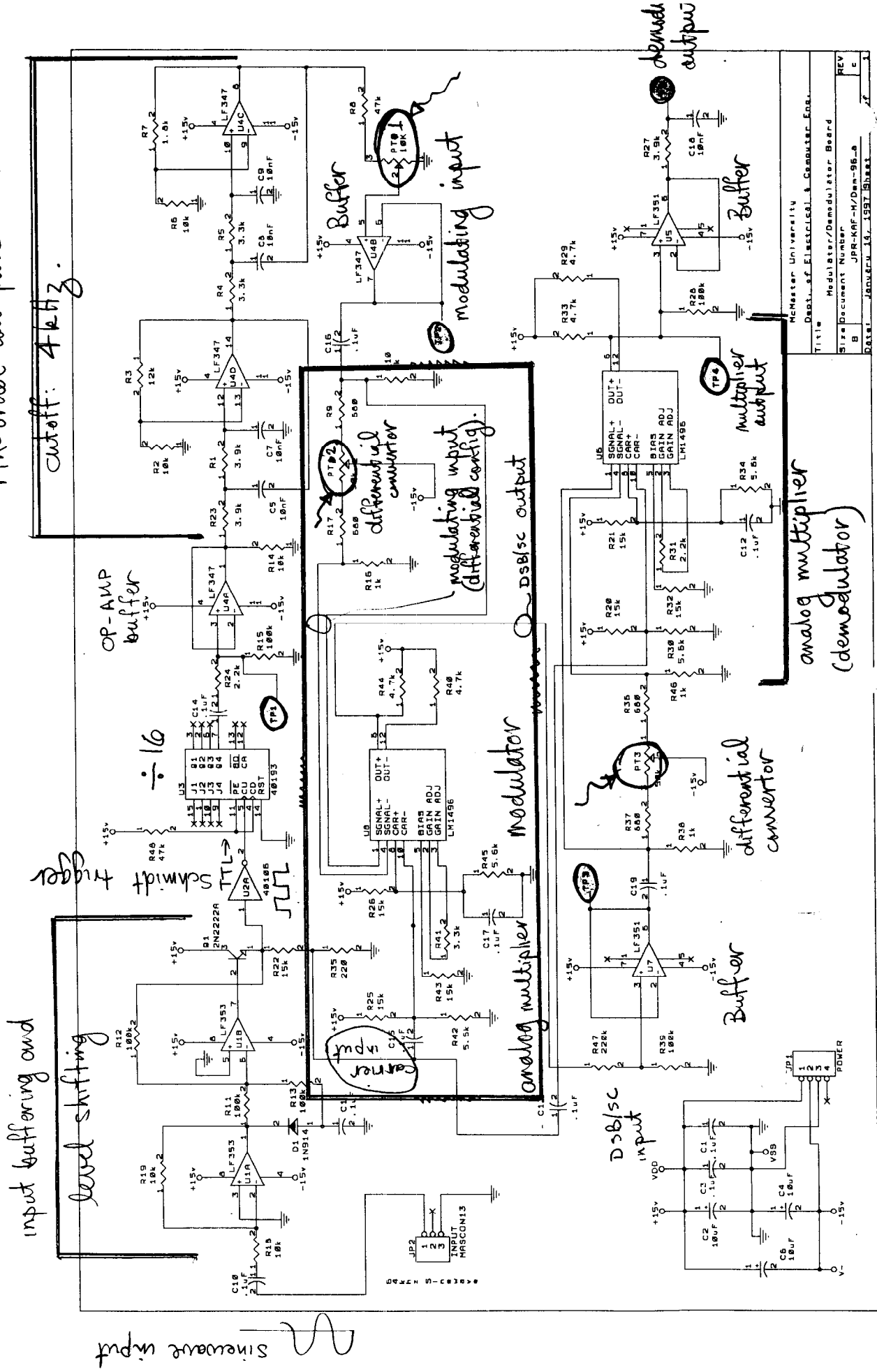


Fig. 1 Block diagram of AM modulator - demodulator circuit

AM Modulator-Demodulator Detailed Schematic

Fig 2

4th-order low-pass Butterworth filter



Lab 3 Report

1. Simulation

Write a matlab code to simulate the DSB-SC modulator-demodulator (MODEM).

Assume that the carrier frequency is 64 kHz and the message is a single tone (sinusoid) at frequency 4 kHz. You may modify the matlab code "dsb_sc_modem.m".

$$f_c = 64 \text{ kHz} \\ f_m = 4 \text{ kHz}$$

(i) Plot the DSB-SC signal both in the time and frequency domains.

(ii) Add a carrier to the DSB-SC signal so that it becomes the standard AM signal.

$$\text{i) } P_c = \frac{1}{2} P_{s0}$$

Consider the following cases: (i) the carrier power is half of the total power in both the sidebands. (ii) the carrier power is three times the total power of the

$$\text{ii) } P_c = 3 P_{s0}$$

sidebands (iii) the carrier power is more than the three times the total power of the sidebands.

$$\text{iii) } P_c > 3 P_{s0}$$

(iii) Multiply the DSB-SC signal (without the carrier addition) by a LO which is phase-synchronized with the carrier, i.e. $\cos(2\pi f_c t)$. Let this be $s'(t)$. Plot $s'(t)$ in both the time and frequency domain

(iv) Let $s'(t)$ pass through an ideal low pass filter which has a cutoff frequency greater than 4 kHz (but less than 124 kHz so that the spectrum around 128 kHz is blocked) and let the output of the LPF be $m'(t)$. Plot $m'(t)$ in both the time and frequency domains.

$$4 \text{ kHz} < \text{LPF} < 124 \text{ kHz}$$

2. Experimental Details

Transmit Section:

(i) Plot the observed DSBSC waveform and the simulated DSB-SC waveform (i.e.

Section 1(i)) in the same figure (in both the time and frequency domains).

Introduce a suitable amplitude scaling factor, if needed. Do you see any discrepancies between the two? Explain.

(ii) Vary the potentiometer PT2 (which adds the carrier) and observe the effect on the signal in both the time and frequency domains. Write a brief description comparing your observed time and frequency domain signals with your matlab plots for the standard AM (i.e. Section 1(ii))

(iii) Explain how the 4 kHz message signal is generated by the board.

Receive Section:

(i) Write a brief description comparing the signal at TP4 with that obtained from the matlab simulations (i.e. 1(iii)) in both the time and frequency domains.

(ii) Compare the demodulated output at TP5 with that obtained from the matlab simulations (i.e. 1(iv)). Comment on the similarity of the demodulated output to the original message $m(t)$. What can be done to improve the similarity?

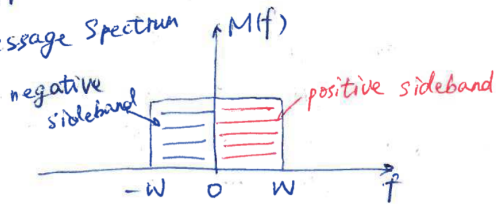
(iii) Describe the observation you have made by varying PT1.

3. Matlab code

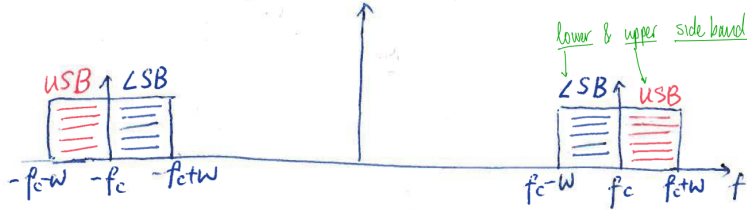
Attach the matlab codes used for simulations.

Upper Sideband (USB) & Lower Sideband (LSB)

Message Spectrum



AM Spectrum



• $P_{USB} : P_{carrier} = ?$, for $1 \text{ mod} = 50\%$ & 100%

• AM wave:

$$m(t) = A_m \cos(2\pi f_m t)$$

$$s(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t)$$

$$= A_c [1 + \frac{k_a A_m}{\mu} \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

$$= A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

$$= A_c \cos(2\pi f_c t) + A_c \mu \cos(2\pi f_m t) \cos(2\pi f_c t)$$

$$= \underbrace{A_c \cos(2\pi f_c t)}_{\text{CARRIER}} + \underbrace{\frac{1}{2} A_c \mu [\cos(2\pi t(f_m - f_c)) + \cos(2\pi t(f_m + f_c))]}_{\text{LSB USB}}$$

• for signal $x(t) = A \cos(2\pi f t)$, mean power of $x(t)$: $\langle p(t) \rangle = \frac{A^2}{2}$

• carrier: $c(t) = A_c \cos(2\pi f_c t)$

↳ power: $P_c = \frac{1}{2} A_c^2$

• USB: $s_{USB}(t) = \frac{1}{2} A_c \mu \cos(2\pi t(f_m + f_c))$

↳ power: $P_{USB} = \frac{1}{2} (\frac{1}{2} A_c \mu)^2$
 $= \frac{1}{2} (\frac{1}{4}) A_c^2 \mu^2$
 $= \frac{1}{8} A_c^2 \mu^2$

• LSB: $s_{LSB}(t) = \frac{1}{2} A_c \mu \cos(2\pi t(f_m - f_c))$

↳ power: $P_{LSB} = \frac{1}{2} (\frac{1}{2} A_c \mu)^2$
 $= \frac{1}{2} (\frac{1}{4}) A_c^2 \mu^2$
 $= \frac{1}{8} A_c^2 \mu^2$

$$\text{ratio: } \frac{P_{USB}}{P_c} = \frac{\frac{1}{8} A_c^2 \mu^2}{\frac{1}{2} A_c^2} = \frac{1}{4} \mu^2$$

$$50\% \text{ mod: } \mu = \frac{1}{2} \rightarrow \frac{1}{4} (\frac{1}{2})^2 = \frac{1}{16}$$

$$100\% \text{ mod: } \mu = 1 \rightarrow \frac{1}{4} (1)^2 = \frac{1}{4}$$

LAB Calcs:

$$P_c = \mu P_{USB}$$

$$\frac{1}{2} A_c^2 = \mu \frac{1}{4} A_c^2 \mu^2$$

$$1 = \mu \frac{1}{2} \mu^2$$

$$\mu^3 = \frac{2}{\mu}$$

$$\mu = \sqrt[3]{\frac{2}{\mu}}$$

$$i) \mu = \sqrt[3]{\frac{2}{\mu}}$$

$$= \sqrt[3]{\frac{2}{\mu}}$$

$$= 2$$

$$\mu = k_a A_m = 2$$

$$\downarrow k_a = 1$$

$$A_m = 2$$

$$ii) \mu = \sqrt[3]{\frac{2}{\mu}} = k_a A_m$$

$$\downarrow$$

$$A_m = \sqrt[3]{\frac{2}{\mu}}$$

$$\% \text{ modulation} = 100\% \times \max |k_a m(t)|$$

$$\max |k_a m(t)| \leq \text{DC bias}$$

$$k_a = \frac{1}{\max |m(t)|} = \frac{1}{1} = k_a = 1$$

from my notes

• successful envelope detection:

• no envelope distortion: $\% \text{ modulation} = 100\% \times \max |k_a m(t)|$
 $\% \text{ modulation} \approx 100\%$

• no freq. overlap & distortion: $f_c > W$

$$\max |k_a m(t)| \leq \text{DC bias}$$

• eg. if $\max |m(t)| = 5V$
 then $k_a = \frac{1}{\max} = \frac{1}{5}$
 then DC bias = $\max |k_a m(t)| = 1$

EE3TR4 Lab 4

Random Processes

Since we are processing signals on a computer in this lab, all the signals must be discrete time. The matlab file **Lab4.m** is posted on Avenue. Please **modify this code to do all the numerical experiments.**

In this lab, we assume all signals are wide-sense stationary, ergodic, and zero mean.

Numerical Experiment #1: Evaluation of Autocorrelation and Power Spectral Density

In this part, we are given a sample of a random process which corresponds to the output $y(n)$ of an ideal low-pass filter (LPF) driven by white noise. The bandwidth of the filter is 250 Hz. The output data $y(n)$ is available in the file “lab4_num_expt1.mat” on Avenue.

Evaluate the autocorrelation sequence corresponding to y , R_y and plot it as a function of the lag τ . Use the matlab command “`xcorr`”, in the form “`Ry=xcorr(y,y,maxlag)`”, where “`maxlag`” is the maximum autocorrelation lag you choose. The default value is the length of y , which is very long, resulting in a very compressed autocorrelation plot. If you use a value, say of 100 or less for `maxlag`, then the plot expands and you can see clearly the features of the autocorrelation function around the lag, $\tau = 0$.

Evaluate the corresponding power spectral density (PSD) and plot it as a function of frequency. As explained in the class, the discrete Fourier transform (DFT) of the autocorrelation is the PSD. To compute the PSD, use “`fftshift(fft(fftshift(Ry)))`” instead of “`fft(Ry)`”. Using the former, the PSD can be easily plotted as a function of the frequency vector. The length of the frequency vector is equal to that of the “`tau_vector`” which is $2*\text{maxlag}+1$. The maximum frequency “`fmax`” is half of the sampling rate ($= 1/\text{tstep}$). Refer to the matlab code “Lab4.m” posted on Avenue.

Because of the way we use the `fft` function, the output will be complex, even though a power spectral density function is supposed to be pure real. This has to do with the way the `fft` input is set up. We can’t go in to the details here, but they will become clear in EE4TL4. To get around this problem, we simply use “`abs(fftshift(fft(fftshift(Ry))))`”, where `abs(.)` evaluates the magnitude of the argument.

Write Up: For this section, (i) calculate the theoretical autocorrelation function of the output, and the corresponding PSD using the methodology discussed in class. Compare (qualitatively) your theoretical results with those from your program, and explain any

discrepancies. Note that the amplitude (i.e. y-axis) of PSD of the matlab could be different from that of the theoretical PSD because of the difference between DFT and Fourier transform.

- (ii) change the maxlag from 100 to 200 and then to 500 and observe its impact on the PSD and comment. Include plots of all relevant quantities.
- (iii) Estimate the bandwidth of the filter using the autocorrelation plot. Hint: measure the locations of zeros of sinc and compare it with the theoretical calculation.

Note: To measure the locations of zeros, use + (data cursor) on matlab plot.

Numerical Experiment #2: A sinusoid buried in noise

A sinusoidal signal of the form

$$x(t) = A \sin(2\pi f_c t + \theta),$$

where θ is a random variable which takes any value between 0 and 2π with equal probability, passes through an AWGN channel of unity gain. The AWGN channel adds white noise.

Download the file “lab4_num_expt2.mat”. It contains a signal $y_t(n)$ which is the output of the AWGN channel in time domain. Estimate the frequency, f_c of the sinusoid.

Write up: For this section, provide the theoretical calculations (i.e. derivations of analytical expressions for autocorrelation and PSD of the AWGN channel output. Refer to lecture notes). Compare the theoretical autocorrelation function and PSD with those from your program (qualitatively).

- (i) Do you observe a peak at the zero lag in the autocorrelation plot? Explain its origin.
- (ii) Change the maxlag from 100 to 200 and then to 20000 and observe its impact on the frequency resolution. Estimating the frequency, f_c in the frequency domain and provide a table of measurements for the maxlag of 100, 200 and 20000. Do the frequency estimates converge? Explain the connection between the maxlag and frequency resolution.
- (iii) Plot y_t as a function of time. It is the sinusoidal signal buried in noise (i.e. $x(t)$ plus white noise). Can you estimate the frequency, f_c directly from y_t , without calculating the correlation? Explain.

Bonus Question: Numerical Experiment #3: Delay Estimation

Download the file “lab4_num_expt3.mat”. A Gaussian pulse given by

$$x(t) = \exp(-t^2)$$

is passed through a channel of transfer function

$$H(f) = \exp(-j2\pi fT),$$

i.e. it simply delays the input by T seconds and adds white noise $w(t)$. The output of the channel is of the form

$$y(t) = \exp[-(t-T)^2] + w(t).$$

The file “lab4_num_expt3.mat” contains the input $x(t)$ as “xt” and the output $y(t)$ as “yt”. Plot $x(t)$ vs time (i.e. “tt” defined in Lab4.m) and $y(t)$ vs time. You will find that you cannot estimate the delay by plotting $y(t)$ since the Gaussian pulse is buried in noise. Develop a technique to estimate the delay T and explain why it works. Provide an estimate of T using your technique.

Hint: Find the cross-correlation between $y(t)$ and $x(t)$ using `xcorr`, i.e.

$$R_{xy}(\tau) = \text{xcorr}(y, x, \text{maxlag})$$

Note that when the $x(t)$ is delayed by $\tau = T$, the correlation between $x(t)$ and $y(t)$ is maximum.

Write Up: For this section, plot $x(t)$ and $y(t)$ vs time. Explain why your technique works and the straightforward approach of finding the delay by plotting $y(t)$ vs time fails. Provide the plot of cross-correlation between x and y . Can you estimate the delay by the autocorrelation of y (i.e. `xcorr(y,y)`)? Explain.

Bonus question is worth 1 marks, i.e. if you finish Lab4 without bonus question, you can get a max. of 5. If you answer bonus question, you can get up to 6 marks.

Lab Report Template:

Do the write ups for each section as explained before. Attach the matlab codes.